



BITS Pilani
Pilani Campus

Digital Design

Prof. GSS Chalapathi,
Prof. Saurabh Gandhi,
Prof. Ramesh Sambangi

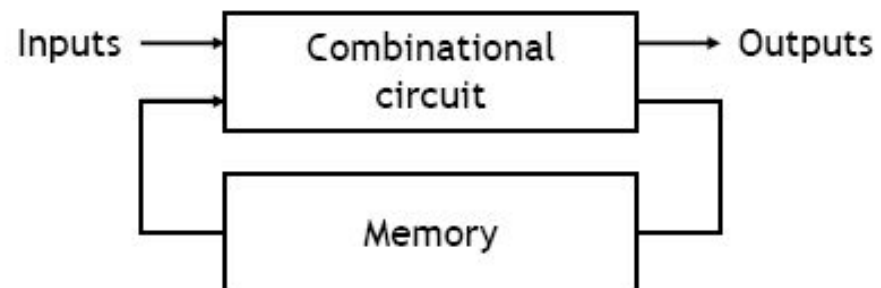
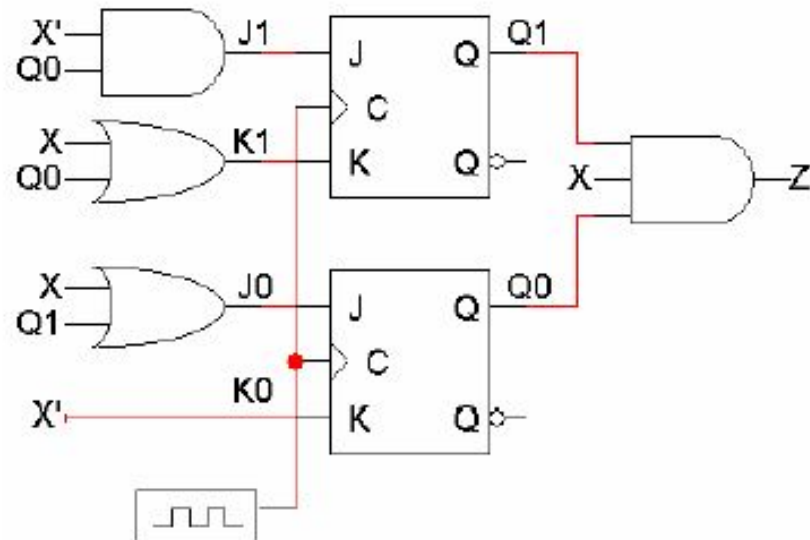
Review questions

1. The table shown below is called a...
2. The characteristic equation for a J-K flip-flop is...
3. The table shown below is called a...
4. The flip-flop shown below will trigger on...

Analyzing sequential circuits



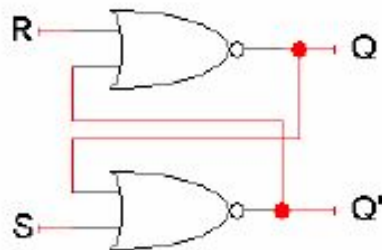
- Here is a sequential circuit with two JK flip-flops. There is one input X and one output Z .
- The values of the flip-flops (Q_1Q_0) form the **state**, or the memory, of the circuit.
- The flip-flop outputs also go back into the primitive gates on the left. This matches the abstract sequential circuit diagram at the bottom.



How do we begin analyzing the circuit?



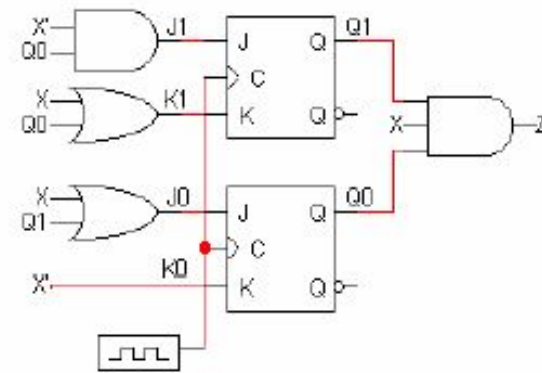
- We can analyze a combinational circuit by deriving a truth table, which shows how the circuit outputs are generated from its inputs.
- But in a sequential circuit, the outputs are dependent upon not only the inputs, but also the current state of the flip-flops. So to understand how a sequential circuit works, we have to know how the memory changes.
- A **state table** is the sequential analog of a truth table. It shows inputs *and* current states on the left, and outputs *and* next states on the right.
- We saw an example of a state table last week, for an SR latch.



Inputs		Current		Next	
S	R	Q	Q'	Q	Q'
0	0	0	1	0	1
0	0	1	0	1	0
0	1	0	1	0	1
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	0	1	0

Analyzing our example circuit

- A state table for our example circuit is shown below.
- The present state Q_1Q_0 and the input X will determine the next state Q_1Q_0 and the output Z .



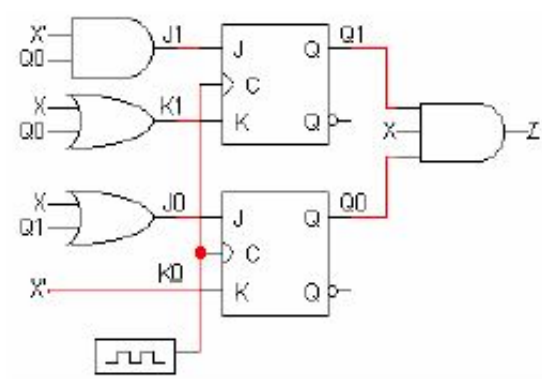
Present State		Inputs X	Next State		Outputs Z
Q_1	Q_0		Q_1	Q_0	
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

Finding the outputs

- From the diagram, you can see that

$Z = Q_1Q_0X$

- This is an example of a **Mealy machine**, where the output depends on both the present state (Q_1Q_0) and the input (X).



Present State		Inputs	Next State		Outputs
Q_1	Q_0	X	Q_1	Q_0	Z
0	0	0			0
0	0	1			0
0	1	0			0
0	1	1			0
1	0	0			0
1	0	1			0
1	1	0			0
1	1	1			1

Finding next states

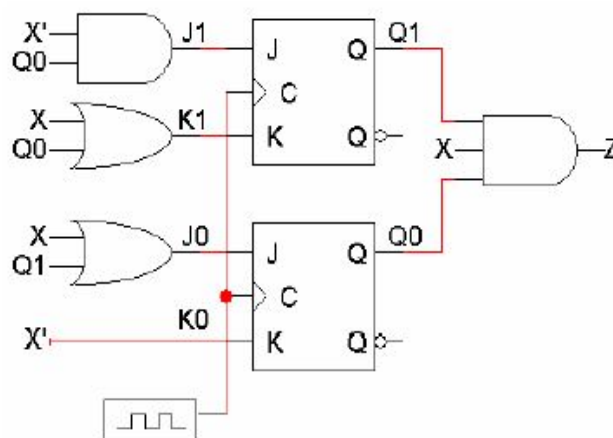


- Finding the next states is harder. To do this, we have to figure out how the flip-flops are changing.
 - Find Boolean expressions for the flip-flop inputs.
 - Use these expressions to find the actual flip-flop input values for each possible combination of present states and inputs.
 - Use flip-flop characteristic tables or equations to find the next states, based on the flip-flop input values and the present states.

- For our example, the **flip-flop input equations** are:

$$\begin{aligned}J_1 &= X' Q_0 \\K_1 &= X + Q_0 \\J_0 &= X + Q_1 \\K_0 &= X'\end{aligned}$$

- JK flip-flops each have two inputs, J and K. (D and T flip-flops have one input each.)



Flip flop input values

- With these equations, we can make a table showing J_1 , K_1 , J_0 and K_0 for the different combinations of present state Q_1Q_0 and input X .

$$J_1 = X' Q_0$$

$$K_1 = X + Q_0$$

$$J_0 = X + Q_1$$

$$K_0 = X'$$

Present State		Inputs X	Flip-flop Inputs			
Q_1	Q_0		J_1	K_1	J_0	K_0
0	0	0	0	0	0	1
0	0	1	0	1	1	0
0	1	0	1	1	0	1
0	1	1	0	1	1	0
1	0	0	0	0	1	1
1	0	1	0	1	1	0
1	1	0	1	1	1	1
1	1	1	0	1	1	0

Finding the next states

- Finally, use the JK flip-flop characteristic tables or equations to find the next state of *each* flip-flop, based on its present state and inputs.
- The general JK flip-flop characteristic equation was given earlier today.

$$Q(t+1) = K'Q(t) + JQ'(t)$$

- In our example circuit, we have two JK flip-flops, so we have to apply this equation to *each* of them.

$$Q_1(t+1) = K_1'Q_1(t) + J_1Q_1'(t)$$

$$Q_0(t+1) = K_0'Q_0(t) + J_0Q_0'(t)$$

Present State		Inputs	Flip-flop Inputs				Next State	
Q_1	Q_0	X	J_1	K_1	J_0	K_0	Q_1	Q_0
0	0	0	0	0	0	1	0	0
0	0	1	0	1	1	0	0	1
0	1	0	1	1	0	1	1	0
0	1	1	0	1	1	0	0	1
1	0	0	0	0	1	1	1	1
1	0	1	0	1	1	0	0	1
1	1	0	1	1	1	1	0	0
1	1	1	0	1	1	0	0	1

State table

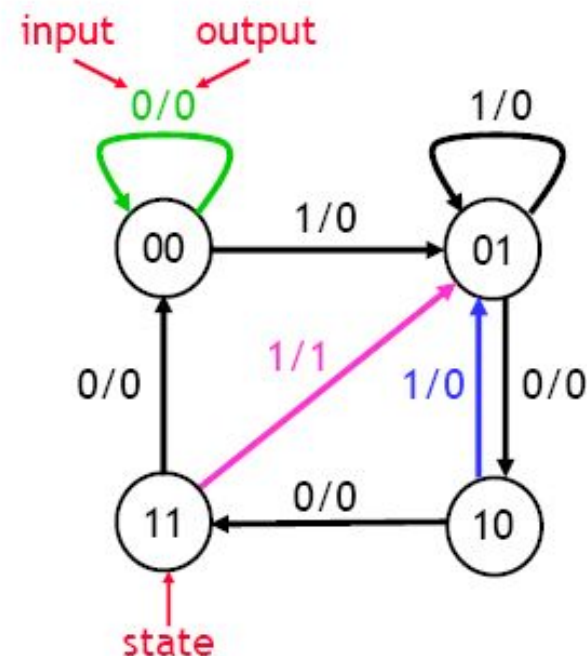
- The table starts with **Present State** and **Inputs**.
 - **Present State** and **Inputs** yield **FF Inputs**.
 - **Present State** and **FF Inputs** yield **Next State**, based on the flip-flop characteristic tables.
 - **Present State** and **Inputs** yield **Output**.
- We really only care about **FF Inputs** in order to find **Next State**.

Present State		Inputs X	Flip-flop Inputs				Next State		Output Z
Q_1	Q_0		J_1	K_1	J_0	K_0	Q_1	Q_0	
0	0	0	0	0	0	1	0	0	0
0	0	1	0	1	1	0	0	1	0
0	1	0	1	1	0	1	1	0	0
0	1	1	0	1	1	0	0	1	0
1	0	0	0	0	1	1	1	1	0
1	0	1	0	1	1	0	0	1	0
1	1	0	1	1	1	1	0	0	0
1	1	1	0	1	1	0	0	1	1

State diagrams

- We can also represent the state table graphically with a **state diagram**.
 - The diagram has one **node** for each possible state.
 - **Arrows** in the diagram connect present states to next states, and are labelled with "input/output."
- A diagram corresponding to our example state table is shown below.

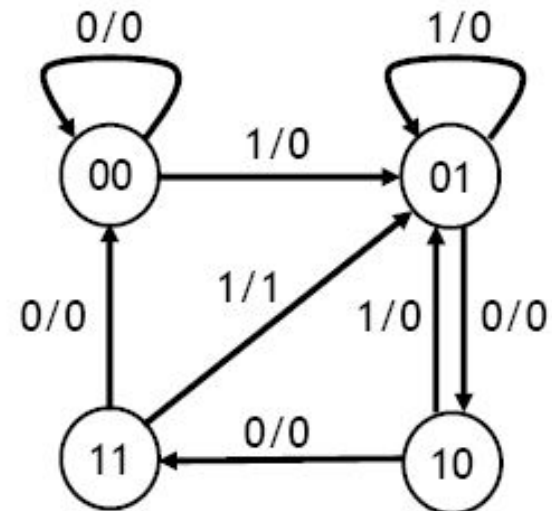
Present State		Inputs X	Next State		Output Z
Q ₁	Q ₀		Q ₁	Q ₀	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	0	1	0
1	1	0	0	0	0
1	1	1	0	1	1



Size of the state diagram

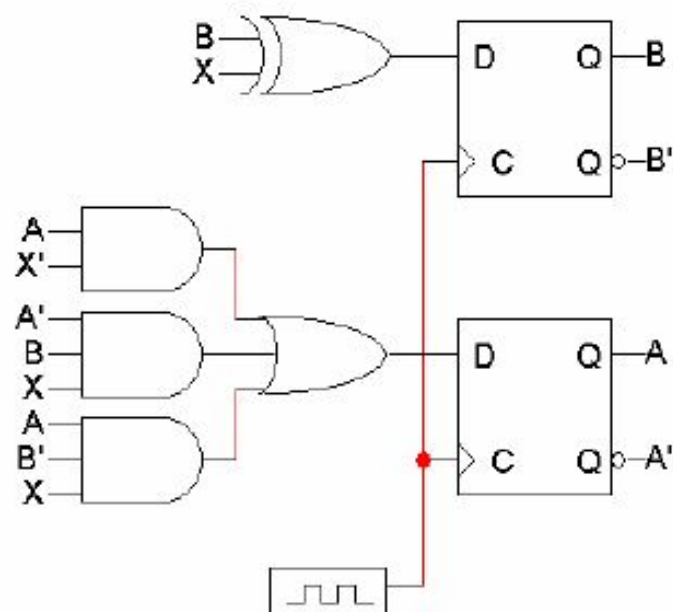
- Always check the size of your state diagrams.
 - If there are n flip-flops, there should be 2^n nodes in the diagram.
 - If there are m inputs, then each node will have 2^m outgoing arrows.
- Our example circuit has two flip-flops and one input, so the state diagram should have four nodes, each with two outgoing arrows.

Present State		Inputs X	Next State		Output Z
Q_1	Q_0		Q_1	Q_0	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	0	1	0
1	1	0	0	0	0
1	1	1	0	1	1



Example

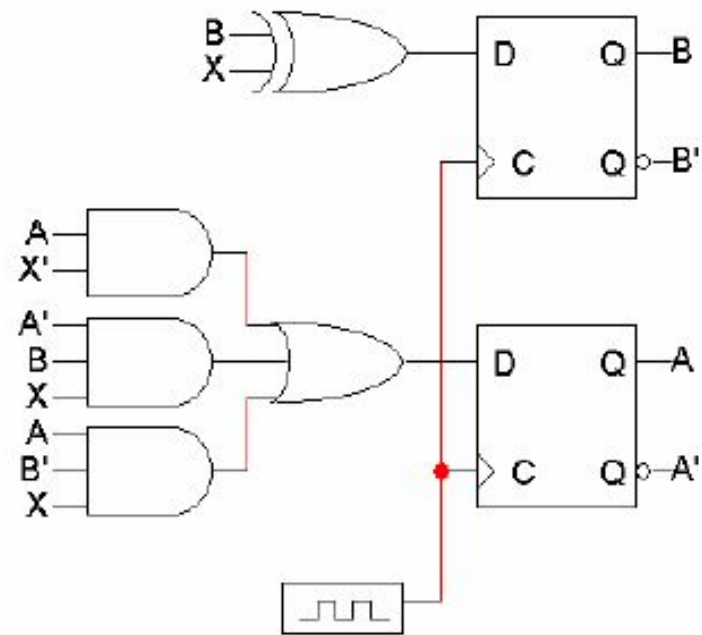
- Here are two D flip-flops, with values labelled **A** and **B**.
- There is one input, **X**.
- There are no explicit outputs.
 - In this case, the outputs are assumed to be the flip-flop values **A** and **B** themselves.
 - This is an example of a **Moore machine**, where the outputs depend on only the present state.



Example – circuit analysis

- The basic state table is below.
- Again, you can see that the present states are being used to generate the next states.
- For this example, remember that the present state also serves as the output.

Present State		Inputs	Next State	
A	B	X	A	B
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		



Step 1: Flip flop input equations

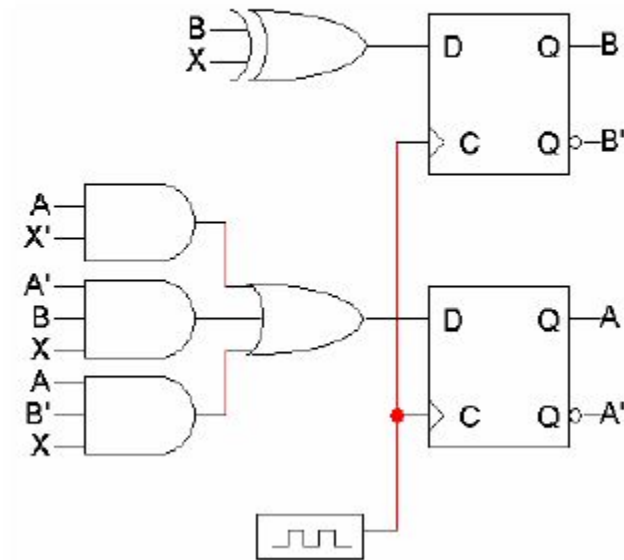


- There are two input equations, one for each flip-flop.

$$D_A = AX' + A'BX + AB'X$$

$$D_B = B \oplus X$$

- "D_A" indicates a D-type flip-flop, whose output is A.



Step 2: Flip flop input values

- Now that we have the equations for D_A and D_B , we can fill in actual values for each combination of present state and inputs.

$$D_A = AX' + A'BX + AB'X$$

$$D_B = B \oplus X$$

Present State		Inputs	Flip-flop Inputs	
A	B	X	D_A	D_B
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	1	0
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

Step 3: finding next states

- Finally, use the D flip-flop characteristic equation to find the next state of each flip-flop, based on its present state and its inputs.
- D flip-flops are simple because the next state is the same as the D input, *regardless* of the present state.

$$Q(t+1) = D$$

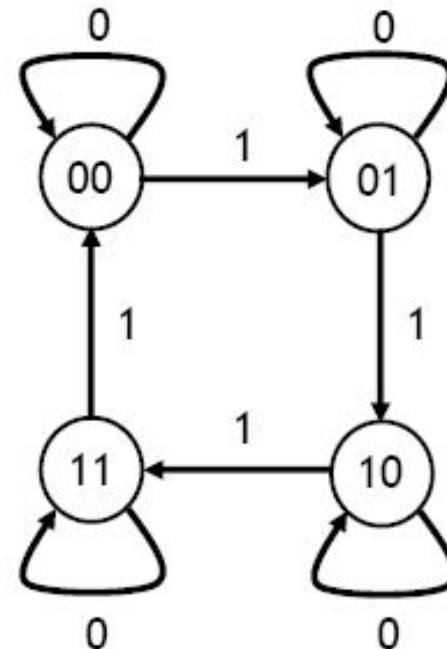
- People often don't even bother writing the "Flip-flop Inputs" columns.

Present State		Inputs	Flip-flop Inputs		Next State	
A	B		D _A	D _B	A	B
0	0	0	0	0	0	0
0	0	1	0	1	0	1
0	1	0	0	1	0	1
0	1	1	1	0	1	0
1	0	0	1	0	1	0
1	0	1	1	1	1	1
1	1	0	1	1	1	1
1	1	1	0	0	0	0

What does this circuit do?

- When $X = 0$, the next state is the same as the present state.
- When $X = 1$, the next state is "one more" than the present state.
- This is a basic two-bit **counter** with an enable input, X . It's also called a modulo-4 counter, since it counts from 0 to 3 repeatedly.

Present State		Inputs	Next State	
A	B		A	B
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	1	0
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0



Summary



- To analyze sequential circuits, you have to understand how the flip-flops change on each clock cycle, according to their current values and inputs.
- A **state table** show all the possible ways that the outputs and state of a sequential circuit can change, based on the its inputs and present state.
- **State diagrams** are an alternative way of showing the same information.
- Next time we'll look at designing sequential circuits. This is the opposite process—you make a state table and/or diagram first, and then turn that into a sequential circuit.